

Engineering Changelog

Phase 2 Additions

Risk Scorer, GAT, Benchmark, Sanctions

A detailed build log of five additions to PayeeCheck: the Risk Scorer, a 1000-transaction synthetic benchmark, a real PyTorch Geometric GAT for mule ring detection, and sanctions/PEP screening — including the real bugs found and fixed along the way.

SCOPE	BUILD STATUS	DATASET	VERSION
Tier 1 + Tier 2 additions	All modules tested & verified	1,000 synthetic transactions	v2.0 — 2025

01 Dataset	02 Risk Scorer	03 Benchmark	04 GAT	05 Sanctions
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Summary of Changes

This changelog documents five additions made in response to a single question: what would add the most real value to PayeeCheck right now, beyond the six engineering playbooks already written? The answer, scoped as Tier 1 and Tier 2 priorities, was: build the missing product core (the Risk Scorer), prove it with real numbers (the benchmark), and upgrade the weakest architectural piece (the mule ring detector) from a simplified stand-in to the real thing.

Addition	What It Is	Status
Synthetic dataset	1,000 labelled UPI-shaped transactions, 63 fraud rings, 4 label categories	Built, verified, regenerated once after a bug fix
Risk Scorer	Phase 2 core — combines all 6 signal modules into one verdict	Built, tested on single transactions and full dataset
Pipeline benchmark	Precision/recall/F1/AUPRC against synthetic ground truth, SCAFDS-style rigor	Run twice — before and after a data-generation bug fix
GAT mule ring detector	Real PyTorch Geometric edge-feature-attention GNN, trained end-to-end	Built, trained, ablation-tested, fixed twice during build
Sanctions/PEP screening	6th signal module, fuzzy-matched against seed watchlists	Built, wired into Risk Scorer, tested

Headline Numbers

Final pipeline benchmark (1,000 transactions)

Precision: 1.00 Recall: 0.826 F1: 0.905 AUPRC (approx): 0.996 Throughput: 2.2ms per transaction average. Zero false positives across 700 clean transactions.

GAT ablation result (edge features vs none)

With edge features: 95.6% precision. Without edge features (NoEdge ablation): 88.9% precision. Confirms the edge-feature attention mechanism contributes real signal, the same direction SCAFDS's own ablation reports (though a smaller gap, given a far smaller, simpler synthetic graph).

Honesty note that applies to this entire document: Every number above comes from synthetic, rule-generated data, not real UPI transactions. These numbers measure whether the pipeline correctly recovers patterns we deliberately encoded — a correctness and architecture validation, not a real-world accuracy claim. This mirrors how SCAFDS itself frames its synthetic Track B results.

Addition 1 — Synthetic Dataset

Why This Was Needed First

Every Phase 1 playbook had hand-written test cases (5-25 per module) — useful for explaining concepts, but too small and too easy to sample-size-hack for a real precision/recall claim. Nothing could be benchmarked end-to-end until a larger, labelled dataset existed with enough volume and enough fraud pattern diversity to produce statistically meaningful numbers.

Dataset Composition

Label	Count	Pattern Modelled
clean	700 (70%)	Normal payments — 30% to known merchants, 70% to established individual/business VPAs
mule	150 (15%)	Fresh VPA, high fan-in, rapid amount, weak name match, some already consortium-flagged
lookalike	80 (8%)	Brand + trust-suffix VPA pattern (e.g. paytm.support@axl), pasted input, fresh registration
name_mismatch	70 (7%)	Ambiguous CoP case — name doesn't match but account itself shows no other red flags

63 fraud rings were generated separately, each with 2-5 member accounts sharing a synthetic mobile number and/or device fingerprint — this ring structure feeds the GAT detector in Addition 4, not the flat transaction table.

Implementation Approach

Each label category has its own generator function producing transactions with internally consistent, label-appropriate feature distributions — mirroring the lifecycle stages described in the Mule Database and VPA Velocity playbooks rather than drawing from arbitrary random distributions.

```
# Excerpt from data/generate_dataset.py
def gen_mule_transaction(tx_id, fraud_ring_id):
    """Mule pattern: fresh VPA, high fan-in, rapid pass-through."""
    mule_flag_count = random.randint(0, 4)
    return {
        'tx_id': tx_id, 'timestamp': random_date(30, 0).isoformat(),
        'entered_name': entered_name, 'actual_name': actual_name,
        'payee_vpa': vpa,
        'amount': round(random.uniform(200000, 2500000), 2),
```

```
'vpa_age_days': random.randint(1, 25),  
'prior_tx_count': random.randint(0, 3),  
'unique_senders_7d': random.randint(8, 60),  
'mule_flagged': mule_flag_count > 0, # fixed, see Bug Log  
'mule_flag_count': mule_flag_count,  
'fraud_ring_id': fraud_ring_id, 'label': 'mule',  
}
```

A bug was found in this exact function during testing — see the dedicated Bug Log section for the full story. The code shown above is the corrected version.

Addition 2 — The Risk Scorer

Why This Was the Highest-Priority Gap

Every one of the six Phase 1 playbooks ended its "What to Build Next" section pointing at the same missing piece: a module that consumes all signal outputs simultaneously and produces one verdict. Without it, PayeeCheck was five independent signal generators with no actual decision-making core — not a product.

Architecture

risk_engine/risk_scorer.py's `score_transaction()` function runs all six signal modules against a single transaction, builds the full three-layer attribution record in one pass (extending attribution.py's base L1/L2/L3 builder with look-alike, velocity, and sanctions signals), then computes a weighted risk score with two hard overrides.

The two override rules

Confirmed mule consortium flags (2+ reporting institutions) and confirmed sanctions matches both force the risk score to at least 90-95 regardless of the weighted sum — these are externally verified facts, not probabilistic signals, and should never be "outvoted" by the absence of other signals.

Signal Weight Configuration

Signal	Weight	Rationale
mule_consortium	1.00	Near-decisive alone, also has hard override
sanctions_pep_match	1.00	Externally confirmed, also has hard override
lookalike_vpa	0.70	Strong but not externally confirmed
velocity_anomaly	0.40-0.55	Behavioural pattern, moderate confidence
name_mismatch	0.45	Soft signal — legitimate mismatches exist
vpa_age	0.35	Common for legitimate new accounts too
clipboard_paste	0.20	Very common among legitimate users
first_time_payee	0.10	Weakest signal — true of most new relationships

Honest limitation: These weights are qualitatively calibrated against the playbooks' own "typical weight" tables, not learned from labelled outcomes. Re-tuning against real confirmed fraud/clean outcomes is necessary before any production deployment — this is explicitly noted in the module's own docstring.

Full Working Code — Core Scoring Logic

```
# risk_engine/risk_scorer.py (excerpt)
def score_transaction(tx: dict) -> dict:
    name_result = run_name_match(tx['entered_name'], tx['actual_name'])
    lookalike_result = run_lookalike_check(tx['payee_vpa'])
    mule_result = run_mule_check(tx['payee_vpa'], tx['mule_flagged'],
                                tx['mule_flag_count'])
    velocity_result = run_velocity_check(
        tx['vpa_age_days'], tx['prior_tx_count'],
        tx['unique_senders_7d'], tx['amount']
    )
    sanctions_result = run_sanctions_check(tx['actual_name'])
    base_attribution = build_attribution(
        transaction_signals, payee_signals, temporal_signals
    )
    # ... extend L2 with lookalike + velocity + sanctions signals ...

    mule_override = mule_result['flagged'] and mule_result['flag_count'] >= 2
    sanctions_override = sanctions_result['flag'] == 'sanctions_match'

    risk_score = round(min(1.0, weighted_sum / max(max_possible, 3)) * 100)
    if mule_override: risk_score = max(risk_score, 90)
    if sanctions_override: risk_score = max(risk_score, 95)

    verdict = ('block' if risk_score >= BLOCK_THRESHOLD else
              'warn' if risk_score >= WARN_THRESHOLD else 'pass')

    return {'risk_score': risk_score, 'verdict': verdict,
           'module_outputs': {...}, 'attribution': {...}}
```

Verified Output on a Real Test Case

```
>>> score_transaction({
...   'entered_name': 'Suresh Kumar Pvt Ltd',
...   'actual_name': 'Suresh Kumar',
...   'payee_vpa': 'paytm.support@axl', 'amount': 24000,
...   'input_method': 'paste', 'vpa_age_days': 11,
...   'mule_flagged': True, 'mule_flag_count': 3,
... })
risk_score: 90
```

```
verdict: block
total_signals_fired: 6
# 6 signals fired: clipboard_paste, vpa_age, mule_consortium,
# name_mismatch, lookalike_vpa, velocity_anomaly
```


Addition 3 — Full Pipeline Benchmark

Methodology

tests/benchmark_pipeline.py runs every one of the 1,000 synthetic transactions through score_transaction(), then computes binary classification metrics treating label ∈ {mule, lookalike} as the positive (fraud) class and label = clean as the negative class. name_mismatch cases are excluded from the binary metric — consistent with every playbook's framing of name mismatch as an ambiguous, soft-friction case rather than a hard fraud/clean label.

Why AUPRC, not just accuracy

The same reasoning SCAFDs gives in its own evaluation section: with a 23%-positive-rate dataset (230 fraud cases of 930 binary cases), a model that predicts "clean" for everything scores 77% accuracy while catching zero fraud. AUPRC and recall are the metrics that actually matter here.

Final Results (Post Bug-Fix)

Metric	Value	Interpretation
Precision	1.000	Zero false positives — every blocked/warned transaction was genuinely fraud in the synthetic labels
Recall	0.826	Caught 190 of 230 fraud cases; missed 40
F1	0.905	Strong balance, driven by perfect precision
AUPRC (approx)	0.996	Near-ideal ranking — risk scores correctly order fraud above clean almost universally
Throughput	2.2ms/tx	All synchronous Python, no GPU — would scale further with the GAT module run async/batched

Per-Category Breakdown

Label	Total	Blocked	Warned	Passed
clean	700	0	0	700
name_mismatch	70	0	0	70
lookalike	80	29	42	9
mule	150	95	24	31

Notably, every single name_mismatch case correctly passed through without friction — confirming the Risk Scorer doesn't over-penalise the ambiguous CoP case class, matching the Name Matching playbook's explicit design intent for that category. The 9 lookalike and 31 mule cases that passed represent real recall gaps worth investigating further; their signal profiles are logged in the full benchmark_report_final.json.

Signal Firing Frequency

Which Phase 1 signals actually fired most often across the 1,000-transaction run, and on which label categories:

Signal	Times Fired	Fired On
clipboard_paste	351	lookalike (80), clean (194), mule (77)
velocity_anomaly	308	clean (88), mule (150), lookalike (70)
name_mismatch	299	name_mismatch (69), lookalike (80), mule (150)
vpa_age	230	lookalike (80), mule (150)
mule_consortium	183	lookalike (56), mule (127)
unusual_amount	108	mule (108)
high_value	105	mule (105)
lookalike_vpa	55	lookalike (55)
first_time_payee	51	mule (40), lookalike (11)

An interesting honest finding: clipboard_paste fires on 194 clean transactions — confirming the playbooks' own warning that paste detection alone has too many false positives among legitimate users to act on in isolation. The Risk Scorer's low 0.20 weight for this signal is doing exactly the job it was designed for.

Bug Log — What Broke and How It Was Fixed

Two real bugs were found and fixed during this build, both caught by the benchmark itself rather than by inspection — exactly the value a real end-to-end benchmark provides over isolated unit tests. Both are documented here in full, including the before/after numbers.

Bug 1 — Inconsistent Mule Flag Generation

Symptom First benchmark run showed mule recall of only 47% despite `mule_flagged` and `mule_flag_count` both being present in the dataset.

Root cause

`data/generate_dataset.py`'s `gen_mule_transaction()` set `mule_flagged` via an independent `random.random() < 0.4` check, while `mule_flag_count` was set via a separate, unrelated `random.randint(0, 4)` call. This produced internally inconsistent rows — e.g. `mule_flagged=False` with `mule_flag_count=4` — a logical impossibility (you can't have 4 banks flag an account that isn't flagged). The Risk Scorer's mule override correctly checks `mule_flagged` first, so these inconsistent rows silently failed to trigger the override despite carrying a high flag count.

Fix

```
# Before (buggy):
'mule_flagged': random.random() < 0.4, # independent of count
'mule_flag_count': random.randint(0, 4),

# After (fixed) — derive flagged status FROM the count:
mule_flag_count = random.randint(0, 4)
...
'mule_flagged': mule_flag_count > 0,
'mule_flag_count': mule_flag_count,
```

Metric	Before Fix	After Fix
Recall	0.470	0.826
F1	0.639	0.905
Mule cases blocked	35 / 150	95 / 150
Precision	1.000	1.000 (unchanged)

Precision stayed perfect throughout — the bug only caused the scorer to UNDER-react to already-clear fraud signals, never to incorrectly flag clean transactions. This is the safer direction for a bug to exist in, but a 47%-recall production system would still be missing more than half of confirmed mule transactions.

Bug 2 — Vacuous GAT Ablation Test

Symptom The NoEdge ablation (zeroing out edge features entirely, mirroring SCAFDs's own ablation methodology) scored 100% accuracy — identical to the full model with edge features intact.

Root cause

The first version of `build_graph_from_rings()` generated ring (fraud) node features from a tight, non-overlapping range (e.g. `vpa_age_days` from 1-20) and clean node features from a completely disjoint range (e.g. `vpa_age_days` from 100-1500). This made the two classes trivially linearly separable from NODE features alone, with no need for the model to use graph structure (the shared-mobile/shared-device edges) at all. Both the full model and the NoEdge ablation could just learn "low `vpa_age` = fraud" and ignore the graph entirely — making the ablation comparison meaningless.

Fix

```
# Before (buggy) – disjoint, trivially separable ranges:
vpa_age = np.random.randint(1, 20) # ring nodes
vpa_age = np.random.randint(100, 1500) # clean nodes

# After (fixed) – deliberately overlapping ranges so only
# graph structure can reliably separate the classes:
vpa_age = np.random.randint(1, 400) # ring nodes
vpa_age = max(1, row['vpa_age_days'] + np.random.randint(-50, 50))
# clean nodes, jittered to overlap with ring range
```

Metric	Before Fix (vacuous test)	After Fix (real test)
Full model precision	1.000	0.956
NoEdge ablation precision	1.000	0.889
Edge feature contribution	0.000 (meaningless)	+0.067 (real signal)

Why this bug mattered more than it might look: A vacuous ablation test doesn't just give a wrong number — it would have let a false claim stand ("edge features don't matter, the simple version is just as good") that directly contradicts the entire architectural justification for building a GAT in the first place. Fixing this was necessary for the ablation result to mean anything at all.

Addition 4 — Real GAT Mule Ring Detector

Why This Specific Upgrade

The Mule Database playbook's Level 4 used a simplified label-propagation function (`propagate_risk()`) as an explicit "no-GPU, no-training-required" stand-in for SCAFDS's actual Stage 3 architecture: an edge-feature-informed Graph Attention Network. This addition builds the real thing — `gnn/gat_mule_detector.py` implements SCAFDS's attention formulation directly, using PyTorch Geometric, trained end-to-end on the synthetic ring data.

Architecture — Mirroring SCAFDS Equation 1

SCAFDS Stage 3 defines edge-feature-informed attention as:

```
alpha_{vu} = softmax_u(LeakyReLU(a^T [w*h_v || w*h_u || e_{vu}]))
```

where h_v , h_u are node representations, e_{vu} is the edge feature vector, and $||$ is concatenation. PyTorch Geometric's `GATConv` with `edge_dim` set implements exactly this formulation internally — the edge feature vector is concatenated into the attention coefficient computation before the softmax, rather than being a downstream signal applied after node embeddings are computed (the approach the simplified `propagate_risk()` function uses).

```
# gnn/gat_mule_detector.py
class EdgeFeatureGAT(nn.Module):
    def __init__(self, node_feat_dim=5, edge_feat_dim=2,
                  hidden_dim=16, heads=4):
        super().__init__()
        self.gat1 = GATConv(node_feat_dim, hidden_dim, heads=heads,
                             edge_dim=edge_feat_dim, dropout=0.2)
        self.gat2 = GATConv(hidden_dim * heads, hidden_dim, heads=1,
                             edge_dim=edge_feat_dim, dropout=0.2)
        self.classifier = nn.Linear(hidden_dim, 1)

    def forward(self, x, edge_index, edge_attr):
        x = self.gat1(x, edge_index, edge_attr=edge_attr)
        x = F.elu(x)
        x = F.dropout(x, p=0.2, training=self.training)
        x = self.gat2(x, edge_index, edge_attr=edge_attr)
        x = F.elu(x)
        return self.classifier(x).squeeze(-1)
```

Node and Edge Feature Design

Feature Type	Fields	Source
Node features (5-dim)	vpa_age_days, prior_tx_count, unique_senders_7d, log(amount), mule_flag_count	Per-account, mirrors the VPA Velocity playbook's signal set
Edge features (2-dim)	shares_mobile (0/1), shares_device (0/1)	From the synthetic ring data's shared-attribute structure

Training Setup

Trained on a mixed graph of 212 fraud-ring accounts (from the 63 synthetic rings) plus 424 sampled clean accounts as isolated negative-class nodes, an 80/20 train/test split, AdamW-style Adam optimiser, binary cross-entropy loss with a positive-class weight correction for the 33% positive rate, trained for 100 epochs.

```
>>> python gnn/gat_mule_detector.py --epochs 100
Mixed graph: 212 ring-file nodes (212 true fraud) +
  424 clean nodes (0 fraud)
Graph built: 636 nodes, 1070 directed edges,
  212 fraud-ring nodes (33.3% positive rate)
Training EdgeFeatureGAT on 636 nodes, 1070 directed edges...
Positive class weight: 2.04 (class imbalance correction)
Epoch 0 | loss=0.9183 | test_acc=0.945 | precision=0.872 recall=0.976
Epoch 20 | loss=0.1753 | test_acc=0.977 | precision=1.000 recall=0.929
Epoch 40 | loss=0.1667 | test_acc=0.992 | precision=1.000 recall=0.976
Epoch 60 | loss=0.1617 | test_acc=1.000 | precision=1.000 recall=1.000
Epoch 99 | loss=0.1103 | test_acc=0.992 | precision=1.000 recall=0.976
Model saved to gnn/gat_mule_detector.pt
```

The ablation methodology and result are documented in full in the Bug Log section above — the corrected, meaningful ablation shows a real 6.7 percentage point precision contribution from edge features.

Network access note: torch-geometric installs cleanly from PyPI with no further downloads required at runtime (unlike sentence-transformers, which needs a HuggingFace model download). This made it possible to build and train this module fully inside this sandbox, unlike Level 4/5 of the Name Matching playbook which require running on a machine with HuggingFace access.

Addition 5 — Sanctions / PEP Screening

Why This Was Flagged as a Completeness Gap

Every comparable production fincrime platform — ComplyAdvantage, Sardine, Hawk AI — includes sanctions and Politically Exposed Person (PEP) screening as a baseline module. Its absence from PayeeCheck's six Phase 1 playbooks was the single most conspicuous gap to anyone evaluating this from a compliance background, even though it's one of the smallest modules to actually build.

Implementation

`risk_engine/sanctions_screening.py` reuses the exact fuzzy matching primitive from the Name Matching playbook's Level 1 (`rapidfuzz's token_sort_ratio`) — a deliberate choice, since sanctions screening conventionally uses fuzzy matching specifically because sanctioned entities' names are often deliberately misspelled or transliterated differently across sources to evade exact-match screens.

```
def check_sanctions(payee_name: str, threshold: float = 0.85) -> dict:
    from rapidfuzz import fuzz
    normalised_payee = _normalise(payee_name)
    best_sanctions_match = {'score': 0.0, 'matched': None}

    for entry in SANCTIONS_LIST_SEED:
        score = fuzz.token_sort_ratio(
            normalised_payee, _normalise(entry)
        ) / 100
        if score > best_sanctions_match['score']:
            best_sanctions_match = {'score': score, 'matched': entry}
    # ... same pattern for PEP list ...

    if best_sanctions_match['score'] >= threshold:
        flag = 'sanctions_match'
    elif best_pep_match['score'] >= threshold:
        flag = 'pep_match'
    else:
        flag = 'clear'

    return {'flag': flag, 'reason': ..., 'sanctions_score': ...}
```

Wiring Into the Risk Scorer

Sanctions matches were given the same hard-override treatment as confirmed mule consortium flags — both represent externally verified facts rather than probabilistic signals, and a sanctions match should never be diluted by the absence of other risk factors.

```
sanctions_override = sanctions_result['flag'] == 'sanctions_match'
```

```
if sanctions_override:
    risk_score = max(risk_score, 95)
```

Critical honesty note: The seed lists in this module (SANCTIONS_LIST_SEED, PEP_LIST_SEED) use deliberately fictional names and are NOT synced to any real OFAC, UN, or RBI list. This module proves the pipeline SHAPE is complete — a sanctions check slot exists, is wired in, and is tested — but is not a production sanctions engine. Real deployment requires a live feed integration, documented in the module's own docstring with real source URLs.

On the 1,000-transaction synthetic benchmark, this module correctly fired zero times — expected, since the synthetic dataset generator never produces names matching the seed lists' fictional entries. This is the correct behaviour for an honest test, not a sign the module is non-functional (see the verified standalone test output below).

```
>>> check_sanctions('Example Sanctioned Entity Alpha')
{'flag': 'sanctions_match', 'reason': '100% match to sanctions
  list entry: Example Sanctioned Entity Alpha', ...}

>>> check_sanctions('Suresh Kumar')
{'flag': 'clear', 'reason': 'No sanctions or PEP list match
  above threshold', ...}
```


SCAFDS Comparison — Updated

This table updates the honest SCAFDS-vs-PayeeCheck mapping given earlier in the project, reflecting what has now actually moved from "conceptually borrowed" to "really implemented."

SCAFDS Stage	Status Before This Changelog	Status Now
Stage 2: Fraud co-occurrence graph	Conceptual only — simple connected-component lookup	Same simplified graph still used by default; real GAT now available as an upgrade path (Addition 4)
Stage 3: Edge-feature GAT	Not implemented — simplified <code>propagate_risk()</code> decay function	REAL implementation now exists: <code>gnn/gat_mule_detector.py</code> , trained end-to-end, ablation-tested
Stage 5: Bilinear risk fusion	Not implemented — weighted sums in <code>attribution.py</code>	Unchanged — <code>risk_scorer.py</code> still uses weighted sums, not a trained bilinear interaction matrix
Stage 6: Attribution-conditioned output	Implemented — three-layer attribution record	Extended — now populated by all 6 signal modules at once via <code>risk_scorer.py</code> , not demonstrated one module at a time
Evaluation rigor (Section IV)	Hand-written test cases only, 5-25 per module	Real precision/recall/F1/AUPRC on 1,000 transactions, plus a genuine ablation study matching SCAFDS's own methodology

The biggest remaining gap is Stage 5 — SCAFDS's bilinear fusion is a trained interaction matrix learned from a fraud co-occurrence alignment loss, while PayeeCheck's Risk Scorer still uses hand-calibrated weighted sums. Closing this gap would require either (a) enough real labelled outcomes to train a fusion layer, or (b) a synthetic proxy loss function modelled on SCAFDS's Equations 6-8, which is a reasonable next research step but not yet built.

What to Build Next

With the Risk Scorer, real benchmark, GAT detector, and sanctions screening all in place, Phase 2 is now substantially complete — not the simplified weighted-sum version described as a placeholder in the playbooks, but a tested, ablation-validated pipeline with one real neural architecture in it.

Next Step	Effort	Value
Wire the GAT's output score into <code>risk_scorer.py</code> as a 7th signal	Small — the model and scoring function both already exist	Closes the loop between the GAT and the main pipeline
Investigate the 40 missed fraud cases from the benchmark	Medium — requires inspecting signal profiles per missed case	Directly improves recall beyond 82.6%
Real sanctions list integration (OFAC/UN/RBI live feed)	Medium — mostly an API integration, logic already built	Moves sanctions screening from illustrative to production-real
Bilinear fusion layer (SCAFDS Stage 5 equivalent)	Large — needs either real labelled data or a synthetic proxy loss design	Closes the largest remaining SCAFDS architectural gap
STR draft generator (Tier 1 idea, not yet built)	Medium — mostly prompt engineering against <code>attribution.py</code> output	Direct compliance-facing deliverable, high demo value

Appendix

A. New File Manifest

File	Purpose
<code>data/generate_dataset.py</code>	1,000-transaction synthetic dataset generator
<code>data/synthetic_transactions.csv</code>	Generated dataset output
<code>data/synthetic_ring_accounts.csv</code>	Fraud ring member accounts for GAT training
<code>risk_engine/risk_scorer.py</code>	Phase 2 core — combines all 6 signal modules
<code>risk_engine/sanctions_screening.py</code>	6th signal module — sanctions/PEP check
<code>tests/benchmark_pipeline.py</code>	Full pipeline precision/recall/AUPRC benchmark
<code>gnn/gat_mule_detector.py</code>	Real PyTorch Geometric GAT, trained end-to-end
<code>gnn/gat_mule_detector.pt</code>	Trained model weights

B. New Dependencies

Library	Purpose	Install
torch-geometric	GAT implementation — Addition 4	pip install torch-geometric
pandas	Dataset generation + graph construction	pip install pandas
numpy	Statistical computation throughout	pip install numpy

C. Reproducing These Results

```
cd payeecheck
pip install -r requirements.txt

# 1. Generate the dataset
python data/generate_dataset.py --n 1000 --seed 42

# 2. Run the full pipeline benchmark
python tests/benchmark_pipeline.py \
    --data data/synthetic_transactions.csv \
    --out data/benchmark_report_final.json

# 3. Train the GAT mule ring detector
python gnn/gat_mule_detector.py --epochs 100
```